



POS / Loyalty Interface Message Specification for Schema V1.0

**Version 1.9
May 17, 2018**

Notice – This document contains proprietary information of Gilbarco Inc. In consideration of the receipt of this document, the recipient agrees not to reproduce, copy, use, or transmit this document and/or the information contained herein unless given advance written permission by Gilbarco Inc. Recipient further agrees to surrender upon Gilbarco's request all Gilbarco proprietary information.

Revision History

Version	Date	Author	Reason/Change
0.1	10/10/06	Howie Berinson	Initial draft
0.2	10/17/06	Howie Berinson	Updates to reflect internal feedback, add field-level details, and improve formatting.
0.3	10/25/06	Howie Berinson	Added KioskID to BeginCustomerRequest and TransactionHeader Changed CancelRedemptionRequest to include LoyaltyRewardID instead of TransactionData Added CustomerID to GetCustomerMessagingRequest and made it optional on FinalizeRewardsRequest Clarified definition of TransactionLine status Added status attribute to Promotion and Discount Added LoyaltyRewardID to TenderInfo Clarified amounts as positive or negative Added LoyaltyReportLocation to the EndPeriodResponse and removed the EndPeriodComplete request and response messages Added Message Flow section with diagrams
1.0	12/5/06	Howie Berinson	Added section on Binary Message Header Replaced CustomerID with LoyaltyID in all messages where it was used Changed GetRewardsRequest to make LoyaltyID & TransactionData optional Changed LoyaltyID to signify a Primary Account Number (PAN) instead of track 2 data if the entryMethod = swipe Clarified use of RewardActions Changed RewardReceiptDescription maxLength to 24 Changed RewardPromptText to 2 tags for short and long versions of the prompt Clarified use of RewardTargetLineNumber Clarified use of RewardDiscountMethod Clarified use of RewardLimit Clarified use of ReceiptData Changed TransactionHeader.TillID maxLength to 16 Clarified use of BusinessPeriod.EndDate & EndTime Changed ReceiptLine barcode attribute to remove values gtin and rss14 Changed ItemLine & MerchandiseCodeLine to add a discountable attribute Changed ItemLine.EntryMethod to remove value PLU

			Changed FuelLine.PriceTierCode & TimeTierCode to be optional Changed FuelLine.ActualSalesPrice to allow a zero value Changed EndPeriodRequest to remove the summary total tags Changed GetLoyaltyOnlineStatusResponse to replace the AllowAccrual and AllowRedemption flags with a PromptForLoyalty flag
1.1	12/6/06	Howie Berinson	Changed GetRewardsResponse.FuelRewards to PrepayFuelRewards to clarify its use Clarified use of RewardTargetLineNumber to reflect use on postpay fuel rewards Clarified use of RewardDiscountMethod to reflect prepay vs. postpay fuel rewards Replaced RewardReceiptDescription with RewardReceiptDescShort and RewardReceiptDescLong Clarified use of GetCustomerMessagingRequest Updated message flows 3.1, 3.2 and 3.5 to reflect POS sending EndCustomer after FinalizeRewards or GetCustomerMessaging Updated message flow 3.2 to reflect OPT sending BeginCustomer and EndCustomer
1.2	12/8/06	Howie Berinson	Changed GetRewardsResponse to make LoyaltyIDValidFlag optional
1.3	2/19/07	Howie Berinson	Changed PrepayFuelReward.RewardLimit to make it mandatory
1.4	4/26/07	Howie Berinson	Changed LoyaltyIDValidFlag to include a string (maxlength 40) to allow the loyalty system to supply a masked loyalty ID for POS reporting
1.5	1/16/09	Ron Klinc	Add optional tone attribute to DisplayCommand Add CycleDuringFueling value to the sequence attribute to DisplayCommand Add optional elements Track1 and Track2 to GetRewardsRequest. Add optional element ProductCode to ItemLine, MerchandiseCodeLine and FuelLine. Add optional element ReceiptLineShort to ReceiptData section of CustomerMessageData. This is to be used for outside transactions only.
1.6	4/12/10	Ron Klinc	Add canUseAsPayment attribute to the LoyaltyIDValidFlag tag. This is used for outside transactions only. Add BeforeFuelingBeforeMerchandise value

			to the sequence attribute to DisplayCommand.
1.7	6/16/14	Lisa Hester	Adding Version 1.0 to title of message spec.
1.8	7/24/15	Lisa Hester	Removed the track1 and track2 tags from the getrewardrequest message. Also removed link to sample crc32 as these links seem to change or move. A user can look for samples on the internet.
1.9	5/17/2018	B. Cannon	Removed extra line feeds throughout. 2.13.1 RequestHeader : correct max length to 12

Table of Contents

1	Introduction.....	7
1.1	Purpose.....	7
1.2	References.....	7
1.3	Terms.....	7
2	Message Formats.....	8
2.1	General Information.....	8
2.2	Binary Message Header	8
2.2.1	Format	8
2.2.2	Checksum Algorithm	9
2.3	BeginCustomer.....	10
2.3.1	Request Message	10
2.4	EndCustomer.....	10
2.4.1	Request Message	10
2.5	GetRewards.....	11
2.5.1	Request Message	11
2.5.2	Response Message.....	11
2.6	CancelRedemption	13
2.6.1	Request Message	13
2.7	GetCustomerMessaging	13
2.7.1	Request Message	13
2.7.2	Response Message.....	13
2.8	FinalizeRewards	14
2.8.1	Request Message	14
2.8.2	Response Message.....	14
2.9	CancelTransaction.....	14
2.9.1	Request Message	14
2.9.2	Response Message.....	14
2.10	GetRewardStatus	14
2.10.1	Request Message	14
2.10.2	Response Message	15
2.11	EndPeriod	15
2.11.1	Request Message	15
2.11.2	Response Message	15
2.12	GetLoyaltyOnlineStatus	15
2.12.1	Request Message	15
2.12.2	Response Message	16
2.13	Common Message Sections.....	17
2.13.1	RequestHeader	17
2.13.2	ResponseHeader	17
2.13.3	CustomerMessageData	17
2.13.4	TransactionData.....	18
2.13.5	TransactionHeader.....	19
2.13.6	BusinessPeriod	20
2.13.7	TransactionLine.....	20
2.13.8	ItemLine	21
2.13.9	ItemCode.....	22
2.13.10	PriceOverride	23
2.13.11	MerchandiseCodeLine	23
2.13.12	FuelLine	24
2.13.13	Promotion.....	25
2.13.14	Discount	26
2.13.15	ItemTax	26
2.13.16	TransactionTax	26

2.13.17	TenderInfo.....	27
2.14	Common Data Element Definitions	28
3	Message Flows.....	31
3.1	Inside Sale	31
3.2	Prepay Fuel Sale	32
3.3	Prepay Underrun	33
3.4	Outside Sale Paid Outside.....	34
3.5	Outside Sale Paid Inside	35
3.6	Outside Attended Sale	36
3.7	Loyalty Status Request.....	37
3.8	End of Period	37
4	Tag Index.....	38

1 Introduction

1.1 Purpose

The purpose of this document is to describe the format and content of a message set to be used as a generic interface between a Retail Petroleum Point of Sale System and a Loyalty Reward System.

1.2 References

POS / Loyalty Interface Use Case Model, Version 1.4, 12/8/06

POS Loyalty 1.2 XML Schema Definition files:

PLXML-POS LoyaltyInterface.xsd
 PLXML-PLIUtility.xsd
 PLXML-PLIDictionary.xsd
 PLXML-PLBaseTypes.xsd

PCATS POS Back Office 3.4 XML Schema Definition files:

NAXML-PBIUtility34.xsd
 NAXML-PBIDictionary34.xsd
 NAXML-PBIBaseTypes34.xsd
 NAXML-PaymentSystemProductCodes34.xsd
 NAXML-PBIUnits34.xsd
 NAXML-LanguageCodes34.xsd
 NAXML-StatesProvinces34.xsd
 DR_CurrencyCode_Full.xsd

1.3 Terms

Term	Meaning
Loyalty Identifier	A token that uniquely identifies a Customer to the Loyalty System. The loyalty identifier can be a magnetic stripe card read via card reader, an RFID tag read via RFID reader, a barcode read via scanner, or a numeric value manually entered by the Cashier at a POS or by the Customer at an OPT.
Loyalty Reward	A loyalty reward can represent points earned toward future discounts, cents off per gallon of fuel, or a dollar amount off an item or transaction.
PPU	Price Per Unit

2 Message Formats

2.1 General Information

The POS Loyalty Interface message set is comprised of a series of request/response messages. In all cases, the request message is sent from the POS System to the Loyalty System, and the response message is sent from the Loyalty System to the POS System. The messages are communicated via TCP/IP Sockets. With the exception of a small binary packet header, the message content is in XML format. The data definitions borrow heavily from the PCATS POS Back Office XML Standard version 3.4.

The following conventions are used in the tables below that describe the request and response messages:

- Tag - the XML tag used to describe an element in the message
- Occurs - the number of times the XML tag may occur in the structure
 - 1: the tag occurs exactly once
 - 0..1: the tag may be absent or may occur one time
 - 0..*: the tag may occur zero or more times, unlimited
 - 1..*: the tag may occur one or more times, unlimited
- Attribute - an attribute used with the tag
- Values - the enumerated values of a tag or attribute

2.2 Binary Message Header

The binary message header must be sent as the first part of each request and response message.

2.2.1 Format

The message header is always 28 bytes long. The format of the header is as follows:

Field	# Bytes	Description
Signature	12	A fixed signature equal to the string POSLOYALTY followed by two nulls
Action	4	1 = Message 2 = Heartbeat
DataLength	4	The length of the XML portion of the message. This value may be greater than or equal to zero.
ChecksumData	4	A checksum on the XML portion of the message. Calculated using the standard CRC-32 algorithm.
ChecksumHeader	4	A checksum on the binary header covering the fields Signature, Action, DataLength, and CheckSumData. Calculated using the standard CRC-32 algorithm.

Here is a sample of the XML content for a message:

```
<BeginCustomerRequest>

<RequestHeader><POS LoyaltyInterfaceVersion>1.0</POS LoyaltyInterfaceVersion><VendorName>Gilbarco</VendorName><VendorModelVersion>Passport V6.00.XX</VendorModelVersion><POSSequenceID>POSSequenceID</POSSequenceID><LoyaltySequenceID>LoyaltySeqID</LoyaltySequenceID><StoreLocationID>12345</StoreLocationID><LoyaltyOfflineFlag value="no"/></RequestHeader><OutsideSalesFlag value="no"/>
  <RegisterID>1</RegisterID>
</BeginCustomerRequest>
```

In this case:

```
Action = 1 (Message)
DataLength = 0x01CC (460 bytes)
ChecksumData = 0x71BF8827
ChecksumHeader = 0x26BC1479
```

The binary header for this sample message would be as follows:

```
|-----Signature-----| |--Action-| |--DataLen-| |--ChkData-| |--ChkHead-|
50 4F 53 4C 4F 59 41 4C 54 59 00 00 01 00 00 00 CC 01 00 00 27 88 BF 71 79 14 BC 26
```

The XML data block following the header would be as follows (in binary form):

```
|-----Data Block-----|
3C 42 65 67 69 6E 43 75 73 74 6F 6D 65 72 52 65 71 75 65 73 74 3E 0D 0A 09 3C 52 65
71 75 65 73 74 48 65 61 64 65 72 3E 3C 50 4F 53 4C 6F 79 61 6C 74 79 49 6E 74 65 72
66 61 63 65 56 65 72 73 69 6F 6E 3E 31 2E 30 3C 2F 50 4F 53 4C 6F 79 61 6C 74 79 49
6E 74 65 72 66 61 63 65 56 65 72 73 69 6F 6E 3E 3C 56 65 6E 64 6F 72 4E 61 6D 65 3E
47 69 6C 62 61 72 63 6F 3C 2F 56 65 6E 64 6F 72 4E 61 6D 65 3E 3C 56 65 6E 64 6F 72
4D 6F 64 65 6C 56 65 72 73 69 6F 6E 3E 50 61 73 73 70 6F 72 74 20 56 36 2E 30 30 2E
58 58 3C 2F 56 65 6E 64 6F 72 4D 6F 64 65 6C 56 65 72 73 69 6F 6E 3E 3C 50 4F 53 53
65 71 75 65 6E 63 65 49 44 3E 50 4F 53 53 65 71 75 65 6E 63 65 49 44 3C 2F 50 4F 53
53 65 71 75 65 6E 63 65 49 44 3E 3C 4C 6F 79 61 6C 74 79 53 65 71 75 65 6E 63 65 49
44 3E 4C 6F 79 61 6C 74 79 53 65 71 49 44 3C 2F 4C 6F 79 61 6C 74 79 53 65 71 75 65
6E 63 65 49 44 3E 3C 53 74 6F 72 65 4C 6F 63 61 74 69 6F 6E 49 44 3E 31 32 33 34 35
3C 2F 53 74 6F 72 65 4C 6F 63 61 74 69 6F 6E 49 44 3E 3C 4C 6F 79 61 6C 74 79 4F 66
66 6C 69 6E 65 46 6C 61 67 20 76 61 6C 75 65 3D 22 6E 6F 22 2F 3E 3C 2F 52 65 71 75
65 73 74 48 65 61 64 65 72 3E 3C 4F 75 74 73 69 64 65 53 61 6C 65 73 46 6C 61 67 20
76 61 6C 75 65 3D 22 6E 6F 22 2F 3E 0D 0A 09 3C 52 65 67 69 73 74 65 72 49 44 3E 31
3C 2F 52 65 67 69 73 74 65 72 49 44 3E 0D 0A 3C 2F 42 65 67 69 6E 43 75 73 74 6F 6D
65 72 52 65 71 75 65 73 74 3E 0D 0A
```

2.2.2 Checksum Algorithm

Calculation of the CheckSumData and CheckSumHeader fields must be done using the standard CRC-32 algorithm as described at <http://www.relisoft.com/Science/CrcMath.html> and http://en.wikipedia.org/wiki/Cyclic_redundancy_check. A sample implementation of this algorithm in Visual Basic can be found at <http://www.vbaccelerator.com/home/vb/Code/Libraries/CRC32/article.asp>.

2.3 BeginCustomer

Used to inform the Loyalty System that a Customer is beginning a transaction. This is informational only, and no response is required. This may be a trigger for the Loyalty System to begin any direct display or audio messaging to the Customer.

2.3.1 Request Message

Tag	Occurs	Description
BeginCustomerRequest	1	
RequestHeader	1	Message header
OutsideSalesFlag	1	Attribute: value = yes no If the value attribute is "yes", the sale is occurring at the pump island or at an outdoor kiosk. If value is "no", the sale is occurring at an inside register. Default is "yes".
<i>Choice of:</i>		
RegisterID	1	The identification of the register at which the sale is taking place. This is typically the register number. Format: nonNegativeInteger
FuelPositionID	1	The identification of the physical location where fuel is being dispensed. This is typically the pump number. Format: nonNegativeInteger
KioskID	1	The identification of the kiosk that finalized the sales being reported. Format: nonNegativeInteger

2.4 EndCustomer

Used to inform the Loyalty System that a Customer is completing a transaction. This is informational only, and no response is required. This may be a trigger for the Loyalty System to finalize any direct display or audio messaging to the Customer.

2.4.1 Request Message

Tag	Occurs	Description
EndCustomerRequest	1	
Request Header	1	Message header

2.5 GetRewards

Used to determine if a given Customer Loyalty Identifier represents a valid member of the loyalty program. Also used to inform the Loyalty System of all transaction details so far, and to get back in return any potential rewards that may be redeemed as part of the current transaction.

2.5.1 Request Message

Tag	Occurs	Description
GetRewardsRequest	1	
RequestHeader	1	Message header
LoyaltyID	0..1	Data that uniquely identifies a Customer
FuelPriceData	0..1	Included only if this is a fuel sale
FuelPrice	1..*	Price details for a single fuel grade
FuelGradeID	1	The identifier of the fuel grade, specified as a PCATS product code. Format: string, maxLength=8
RegularSellPrice	1	The normal PPU for the fuel grade. Format: decimal
TransactionData	0..1	POS transaction details

2.5.2 Response Message

Tag	Occurs	Description
GetRewardsResponse	1	
ResponseHeader	1	Message header
LoyaltyIDValidFlag	0..1	The loyalty ID string to be used for POS logging and reporting. For example, this can be a masked version of the actual loyalty ID. If empty, the POS will use the actual loyalty ID sent to the loyalty host for reporting. Format: string, maxLength=40 Attribute: value = yes no Indicates if the LoyaltyID supplied by the POS System is recognized as valid by the Loyalty System. Default is "yes". This value must be supplied if the GetRewardsRequest includes a LoyaltyID. Attribute: canUseAsPayment = yes no Indicates if the loyalty ID supplied by the POS System was recognized as a possible payment method. Applies to outside transactions only. Default is "no".
RewardActions	0..1	Details of loyalty rewards that

		should be added to or removed from the transaction. Once a loyalty reward is added to a transaction (through a FuelRewards or AddReward action), the reward will remain in place unless a RemoveReward action is provided to remove it or a new reward with the same LoyaltyRewardID is sent to replace the original one.
PrepayFuelRewards	0..1	Details of fuel rewards that may be redeemed. This applies only to fuel that has not been dispensed. A reward on fuel that has already been dispensed must be provided under AddReward.
LoyaltyRewardID	1	See LoyaltyRewardID
InstantRewardFlag	1	See InstantRewardFlag
RewardPromptTextShort	0..1	See RewardPromptTextShort
RewardPromptTextLong	0..1	See RewardPromptTextLong
RewardDiscountMethod	1	See RewardDiscountMethod
RewardLimit	1	See RewardLimit
RewardReceiptDescShort	1	See RewardReceiptDescShort
RewardReceiptDescLong	1	See RewardReceiptDescLong
FuelGradeReward	1..*	Details of a fuel discount for a specific grade.
FuelGradeID	1	The identifier of the fuel grade, specified as a PCATS product code. Format: string, maxLength=8
RewardValue	1	See RewardValue
AddReward	0..*	Details of a loyalty reward that should be added to the transaction.
LoyaltyRewardID	1	See LoyaltyRewardID
InstantRewardFlag	1	See InstantRewardFlag
RewardPromptTextShort	0..1	See RewardPromptTextShort
RewardPromptTextLong	0..1	See RewardPromptTextLong
RewardTargetLineNumber	1	Transaction line number of the item to be discounted. If this refers to a fuel item it must be a postpay fuel item - otherwise the discount is ignored. If the reward targets a line that already has a loyalty promotion, then if the LoyaltyRewardID on the new reward matches the ID of the existing reward, the POS will replace the existing promotion with the new one. If the ID is different, then both rewards will apply. If the target is 0 (zero), then the reward discount is applied to the entire transaction as a POS-defined loyalty tender.
RewardDiscountMethod	1	See RewardDiscountMethod

RewardValue	1	See RewardValue
RewardLimit	0..1	See RewardLimit
RewardReceiptDescShort	1	See RewardReceiptDescShort
RewardReceiptDescLong	1	See RewardReceiptDescLong
RemoveReward	0..*	Details of a loyalty reward that should be removed from the transaction.
LoyaltyRewardID	1	The identifier of the loyalty reward discount that should be removed. See LoyaltyRewardID
CustomerMessageData	0..1	Display and/or Receipt messages to be provided to the Customer.

2.6 CancelRedemption

Used to inform the Loyalty System that the Customer has chosen not to redeem one or more optional rewards that were offered. This is informational only, and no response is required.

2.6.1 Request Message

Tag	Occurs	Description
CancelRedemptionRequest	1	
RequestHeader	1	Message header
LoyaltyRewardID	1..*	The identifier of the loyalty reward discount the Customer chose not to redeem. See LoyaltyRewardID

2.7 GetCustomerMessaging

Used to inform the Loyalty System of transaction details for a transaction in progress, and to get back in return any display or receipt messages as appropriate.

2.7.1 Request Message

Tag	Occurs	Description
GetCustomerMessagingRequest	1	
RequestHeader	1	Message header
LoyaltyID	0..1	Data that uniquely identifies a Customer. If this tag does not exist then a valid loyalty ID was not presented during the transaction.
TransactionData	1	POS transaction details. Note that the details provided here will reflect a partial transaction (e.g. a prepay that has not been fueled).

2.7.2 Response Message

Tag	Occurs	Description
GetCustomerMessagingResponse	1	
ResponseHeader	1	Message header
CustomerMessageData	0..1	Display and/or Receipt messages to be provided to the Customer.

2.8 FinalizeRewards

Used to inform the Loyalty System of the final transaction details, and to get back in return any display or receipt messages as appropriate. The Loyalty System should commit all Customer accruals and redemptions at this point. If a valid loyalty ID was not presented during the transaction, the Loyalty System can still respond with display and/or receipt text for loyalty recruiting purposes.

2.8.1 Request Message

Tag	Occurs	Description
FinalizeRewardsRequest	1	
RequestHeader	1	Message header
LoyaltyID	0..1	Data that uniquely identifies a Customer. If this tag does not exist then a valid loyalty ID was not presented during the transaction.
TransactionData	1	POS transaction details

2.8.2 Response Message

Tag	Occurs	Description
FinalizeRewardsResponse	1	
ResponseHeader	1	Message header
CustomerMessageData	0..1	Display and/or Receipt messages to be provided to the Customer.

2.9 CancelTransaction

Used to inform the Loyalty System that the Customer or the Cashier has canceled or voided the transaction. The Loyalty System should cancel or roll back any Customer accruals or redemptions pending for the transaction at this point. This is normally an informational message, but a response is required to ensure that a stored and forwarded request has been successfully received.

2.9.1 Request Message

Tag	Occurs	Description
CancelTransactionRequest	1	
RequestHeader	1	Message header

2.9.2 Response Message

Tag	Occurs	Description
CancelTransactionResponse	1	
ResponseHeader	1	Message header

2.10 GetRewardStatus

Used to get a summary or detailed status of a Customer's loyalty account that can be displayed and/or printed at a POS.

2.10.1 Request Message

Tag	Occurs	Description
GetRewardStatusRequest	1	Attribute: rewardStatusType = summary detail

RequestHeader	1	Message header
LoyaltyID	1	Data that uniquely identifies a Customer

2.10.2 Response Message

Tag	Occurs	Description
GetRewardStatusResponse	1	
ResponseHeader	1	Message header
LoyaltyIDValidFlag	1	Attribute: value = yes no Indicates if the LoyaltyID supplied by the POS System is recognized as valid by the Loyalty System. Default is "yes".
CustomerMessageData	1	Display and/or Receipt messages to be provided to the Customer.

2.11 EndPeriod

Used to inform the Loyalty System that the POS System is ending a business period and beginning a new one. The Loyalty System response can include a reference to any HTML period reports generated by the Loyalty System so the reports can be included in the POS System's end of period reporting stream. If the POS System does not receive an EndPeriodResponse by the time it's ready to print the POS period reports, the POS will skip any loyalty reports for that period.

2.11.1 Request Message

Tag	Occurs	Description
EndPeriodRequest	1	
RequestHeader	1	Message header
FromPeriod	1	The business period being closed on the POS System.
BusinessPeriod	1	Business period definition
ToPeriod	1	The business period being opened on the POS System.
BusinessPeriod	1	Business period definition

2.11.2 Response Message

Tag	Occurs	Description
EndPeriodResponse	1	
ResponseHeader	1	Message header
LoyaltyReportLocation	0..1	A URL path to the location of any HTML end of period reports on the Loyalty System. Format: string, maxLength=100

2.12 GetLoyaltyOnlineStatus

Used to confirm that the Loyalty System is online and accessible.

2.12.1 Request Message

Tag	Occurs	Description
GetLoyaltyOnlineStatusRequest		
RequestHeader		Message header

2.12.2 Response Message

Tag	Occurs	Description
GetLoyaltyOnlineStatusResponse	1	
ResponseHeader	1	Message header
PromptForLoyaltyFlag	1	Attribute: value = yes no Indicates if the POS System should prompt for loyalty ID. If the in-store loyalty server is offline to the central loyalty host, the Loyalty System may not be able to process any messages, so this flag allows the Loyalty System to inform the POS System that all loyalty functions are unavailable. Default is "yes".

2.13 Common Message Sections

2.13.1 RequestHeader

Data elements required for any request message.

Tag	Occurs	Description
RequestHeader	1	
POS LoyaltyInterfaceVersion	1	The version number of the POS Loyalty Interface in use by the system that generated the message. Format: string, maxLength=40
VendorName	1	The POS System vendor name. Format: string, maxLength=40
VendorModelVersion	1	The POS System version. Format: string, maxLength=40
POSSequenceID	1	See POSSequenceID
LoyaltySequenceID	1	See LoyaltySequenceID
StoreLocationID	1	The store identifier, as known to the Loyalty System. The POS System will use a configurable option to populate this value. Format: string, maxLength=12
LoyaltyOfflineFlag	1	Attribute: value = yes no Indicates if the POS System was offline to the Loyalty System when the POS transaction occurred. If "yes" then the Loyalty System can assume that the message has been stored and forwarded. Default is "no".

2.13.2 ResponseHeader

Data elements required for any response message.

Tag	Occurs	Description
ResponseHeader	1	
POS LoyaltyInterfaceVersion	1	The version number of the POS Loyalty Interface in use by the system that generated the message. Format: string, maxLength=40
VendorName	1	The Loyalty System vendor name. Format: string, maxLength=40
VendorModelVersion	1	The Loyalty System version. Format: string, maxLength=40
POSSequenceID	1	See POSSequenceID
LoyaltySequenceID	1	See LoyaltySequenceID
ErrorText	0..1	Free format text for use by the Loyalty System to report error details to the POS System. Format: string, maxLength=80

2.13.3 CustomerMessageData

Display and/or Receipt messages to be provided to the Customer.

Tag	Occurs	Description
CustomerMessageData	1	
DisplayData	0..1	Display commands for the POS System.
DisplayCommand	1..*	A command that directs the POS System to display text to the Customer or the Cashier. Attribute: device = OPT POS-Cashier POS-PoleDisplay POS-CustTerminal The device on which the text should be displayed. Attribute: sequence = WhenReceived BeforeFueling BeforeFuelingBeforeMerchandise DuringFueling CycleDuringFueling AfterFueling The time or time period when the text should be displayed. Attribute: duration = The number of seconds to display the text. Format: nonNegativeInteger Attribute (optional): tone = yes no Determines if an audible sound should be made at the OPT.
DisplayLine	1..*	A single line of display text. Format: string, maxLength=40
ReceiptData	0..1	Receipt commands for the POS System. The POS System should include the ReceiptLines on the final Customer receipt above the footer in the order provided. The POS System should include only the last set of ReceiptData sent at the GetCustomerMessagingResponse or FinalizeRewardsResponse.
ReceiptLine	1..*	A single line of receipt text. If the barcodeFormat attribute is present, this line represents a barcode value of the stated format to be included on the receipt. Format: string, maxLength=40 Attribute: barcodeFormat = upcA upcE ean8 ean13
ReceiptLineShort	0..*	A single line of receipt text. Used for outside transactions only. Format: string, maxLength=20

2.13.4 TransactionData

Data elements that describe a POS transaction.

Tag	Occurs	Description
-----	--------	-------------

TransactionData	1	
TransactionHeader	1	Data elements that describe where, when and how a transaction occurred.
TransactionDetailGroup	0..1	If present, the transaction includes one or more lines.
TransactionLine	1..*	Data elements that describe a single line in the transaction.

2.13.5 TransactionHeader

Data elements that describe where, when and how a POS transaction occurred.

Tag	Occurs	Description
TransactionHeader	1	
TrainingModeFlag	0..1	Attribute: value = yes no If the value attribute is "yes" this is a practice transaction performed by a cashier doing training. Default is "no".
OutsideSalesFlag	1	Attribute: value = yes no If the value attribute is "yes" this transaction is occurring at a dispenser or kiosk outside the store. If value is "no" this transaction is occurring at a POS register inside the store. Default is "yes".
<i>Choice of:</i>		
RegisterID	1	The identification of the register at which the sale is taking place. This is typically the register number. Format: nonNegativeInteger
FuelPositionID	1	The identification of the physical location where fuel is being dispensed. This is typically the pump number. Format: nonNegativeInteger
KioskID	1	The identification of the kiosk that finalized the sales being reported. Format: nonNegativeInteger
CashierID	0..1	Identifies the cashier or attendant conducting the transaction, if any. This value would not apply to a Customer-driven transaction at an OPT or Kiosk. Format: string, maxLength=40
TillID	1	The identifier of the POS System till with financial accountability for the transaction. Format: string, maxLength=16
POSTransactionID	1	The identification of the transaction, as known to the POS System. Format: string, maxLength=16
BusinessPeriod	1	The POS System business period

		within which the transaction takes place.
EventStartDate	1	The start date of the transaction. Format: YYYY-MM-DD
EventStartTime	1	The start time of the transaction. Format: HH:MM:SS
EventEndDate	0..1	The end date of the transaction. Format: YYYY-MM-DD
EventEndTime	0..1	The end time of the transaction. Format: HH:MM:SS

2.13.6 BusinessPeriod

A business period in the POS System.

Tag	Occurs	Description
BusinessPeriod	1	
BusinessDate	1	The business date assigned by the company. It may not be the same as the calendar date. Format: YYYY-MM-DD
PrimaryReportPeriod	1	The major POS System reporting period associated with the data. Format: string, maxLength=12 Attribute: interval = cashier day week month year other
SecondaryReportPeriod	1	The secondary POS System reporting period that is a subset of the major reporting period. If this value is greater than zero; this indicates a shift is closing. If this value is zero; this indicates a day is closing. Format: string, maxLength=40 Attribute: interval = cashier day week month year other
BeginDate	1	The start date of the business period Format: YYYY-MM-DD
BeginTime	1	The start time of the business period Format: HH:MM:SS
EndDate	1	The end date of the business period. If the period is still active this value will be 2100-01-01. Format: YYYY-MM-DD
EndTime	1	The end time of the business period. If the period is still active this value will be 00:00:00. Format: HH:MM:SS

2.13.7 TransactionLine

Data elements that describe a single line item in a POS transaction.

Tag	Occurs	Description
TransactionLine	1	Attribute: status = normal return void

		The status of the transaction line. Normal indicates a normally completed, unmodified transaction line. Return indicates a transaction line for a product returned by the customer for credit as part of a refund or exchange transaction. Return quantity is assumed to be a positive number. Return is not applicable to TransactionTax or TenderInfo transaction lines. Void indicates a transaction line that was rung and then reversed, and is therefore not included in the transaction total and does not impact store inventory.
LineNumber	1	The sequence number assigned to the transaction line by the POS System. Format: nonNegativeInteger
<i>Choice of:</i>		
ItemLine	1	Data elements that describe a non-fuel sale item.
MerchandiseCodeLine	1	Data elements that describe an open department sale item.
FuelLine	1	Data elements that describe a fuel sale item.
TransactionTax	1	Data elements that describe a tax applied to a transaction.
TenderInfo	1	Data elements that describe a tender.

2.13.8 ItemLine

Data elements that describe a non-fuel sale item.

Tag	Occurs	Description
ItemLine	1	Attribute: discountable = yes no Indicates if a discount or promotion may be applied to the item. Default is "yes".
ItemCode	1	Data elements that identify the item.
MerchandiseCode	1	The department or category to which the item belongs. Format: string, maxLength=16
Description	0..1	A description of the item. Format: string, maxLength=40
EntryMethod	1	The method by which an item was identified for the purposes of the transaction. Values: scan manual other
ActualSalesPrice	1	The actual (unit) price at which the item was sold. May be negative. Format: decimal

RegularSellPrice	1	The non-promotional (unit) price of the item. May be negative. It is the price at which the item should normally be sold as modified by the provided qualifiers. Format: decimal
SellingUnits	1	Number of individual units in a transaction item at time of sale. For example, 6 for a 6 pack. Must be greater than zero. Format: decimal
SalesQuantity	1	The quantity sold of the item. Must be greater than zero. Format: decimal
SalesAmount	1	The extended currency amount of the item before tax, not inclusive of promotions or discounts. May be negative. Format: decimal
PriceOverride	0..1	Data elements that describe a Cashier price override.
Promotion	0..*	Data elements that describe a promotional discount automatically applied to the item.
Discount	0..*	Data elements that describe a manual discount applied by the Cashier at a POS register.
ItemTax	0..*	Data elements that describe the prorated amount of a given tax applied to an item.
LinkedFromLineNumber	0..1	The line number of the parent transaction line that caused this item to be added to the transaction. Format: nonNegativeInteger
ProductCode	0..1	The item product code. Format: string

2.13.9 ItemCode

Data elements that identify a non-fuel sale item.

Tag	Occurs	Description
ItemCode	1	
POSCodeFormat	1	The type of coding scheme used in POSCode. Values: upcA upcE ean8 ean13 plu gtin rss14 none
POSCode	1	The scan code or PLU number used to access an item's information. Format: nonNegativeInteger
POSCodeModifier	1	Scan code or PLU modifier. In combination with POSCode and POSCodeFormat it uniquely identifies an item. If it is not used a value of

		"0" (zero) must be provided. Format: nonNegativeInteger Attribute: name A name associated with the modifier. Format: string, maxLength=10
--	--	---

2.13.10 *PriceOverride*

Data elements that describe a Cashier price override.

Tag	Occurs	Description
PriceOverride	1	
PriceOverridePrice	1	The item's unit price after overriding. May be negative. Format: decimal
PriceOverrideReason	1	A reason code indicating why the item/merchandise code/fuel grade was sold at a price different from the RegularSellPrice. Values: as defined by PCATS

2.13.11 *MerchandiseCodeLine*

Data elements that describe an open department sale item.

Tag	Occurs	Description
MerchandiseCodeLine	1	Attribute: discountable = yes no Indicates if a discount or promotion may be applied to the item. Default is "yes".
MerchandiseCode	1	The department or category sold. Format: string, maxLength=16
Description	0..1	A description of the department or category sold. Format: string, maxLength=40
ActualSalesPrice	1	The actual (unit) price at which the item was sold. May be negative. Format: decimal
SalesQuantity	1	The quantity sold of the item. Must be greater than zero. Format: decimal
SalesAmount	1	The extended currency amount of the item before tax, not inclusive of promotions or discounts. May be negative. Format: decimal
Promotion	0..*	Data elements that describe a promotional discount automatically applied to the item.
Discount	0..*	Data elements that describe a manual discount applied by the Cashier at a POS register.
ItemTax	0..*	Data elements that describe the prorated amount of a given tax applied to an item.

ProductCode	0..1	The item product code. Format: string
-------------	------	---------------------------------------

2.13.12 *FuelLine*

Data elements that describe a fuel sale item.

Tag	Occurs	Description
FuelLine	1	Attribute: fuelPrepayFlag = yes no Indicates whether the item represents fuel that has not yet been dispensed.
FuelGradeID	1	The identifier of the fuel grade, specified as a PCATS product code. Format: string, maxLength=8
FuelPositionID	1	The identification of the physical location where fuel is being dispensed. This is typically the pump number. Format: nonNegativeInteger
PriceTierCode	0..1	The type of pricing level by which the customer purchased or may purchase the fuel grade. Format: nonNegativeInteger
TimeTierCode	0..1	The type of time tier by which the customer purchased or may purchase a fuel grade. Format: string, maxLength=16
ServiceLevelCode	1	A code denoting the type of service by which the customer purchases the fuel grade. Values: full self mini
Description	0..1	A description of the fuel grade sold.
ActualSalesPrice	1	The actual PPU at which the fuel grade was sold. Must be greater than or equal to zero. Format: decimal
RegularSellPrice	1	The PPU at which the fuel grade should normally be sold. Must be greater than zero. Format: decimal
SalesQuantity	1	The volume of fuel sold. Must be greater than zero. Format: decimal
SalesAmount	1	The extended currency amount of the fuel, not inclusive of promotions or discounts. Must be greater than zero. Format: decimal
Promotion	0..*	Data elements that describe a promotional discount automatically applied to the fuel.
Discount	0..*	Data elements that describe a manual discount applied by the

		Cashier at a POS register.
ItemTax	0..*	Data elements that describe the prorated amount of a given tax applied to an item.
ProductCode	0..1	The item product code. Format: string

2.13.13 Promotion

Data elements that describe a promotional discount automatically applied to an item or fuel item.

Tag	Occurs	Description
Promotion	1	Attribute: status = normal void The status of the promotion. Normal indicates an active promotion. Void indicates a promotion that is not included in the transaction total.
PromotionID	1	The identifier of the promotion as known to the POS System. Format: string, maxLength=40
LoyaltyRewardID	0..1	The identifier of the promotion as known to the Loyalty System. Applies only where the PromotionReason is "loyaltyOffer". See LoyaltyRewardID
PromotionAmount	1	The amount of the promotion associated with the individual item in the transaction. Must be greater than or equal to zero. This amount is subtracted from the item's SalesAmount. Format: decimal
PromotionReason	1	The reason code for the promotion. Values: combinationOffer mixAndMatchOffer posFuelDiscount posPromotion loyaltyOffer other

2.13.14 Discount

Data elements that describe a manual discount applied by a Cashier at a POS register.

Tag	Occurs	Description
Discount	1	Attribute: status = normal void The status of the discount. Normal indicates an active discount. Void indicates a discount that was rung and then manually reversed, and is therefore not included in the transaction total.
DiscountID	1	The identifier of the discount as known to the POS System. Format: string, maxLength=40
DiscountAmount	1	The amount of the discount associated with the individual item in the transaction. Must be greater than or equal to zero. This amount is subtracted from the item's SalesAmount. Format: decimal
DiscountReason	1	The reason code for the discount. Format: as defined by PCATS

2.13.15 ItemTax

Data elements that describe the prorated amount of a given tax applied to an item.

Tag	Occurs	Description
ItemTax	1	
TaxLevelID	1	The identifier of the tax rate or tax table. Format: string, maxLength=4
<i>Choice of:</i>		
TaxCollectedAmount	1	The amount of this type of tax collected for the item. Must be greater than zero. Format: decimal
TaxRefundedAmount	1	The amount of this type of tax refunded for the item. Must be greater than zero. Format: decimal.

2.13.16 TransactionTax

Data elements that describe a tax applied to a transaction.

Tag	Occurs	Description
TransactionTax	1	
TaxLevelID	1	The identifier of the tax rate or tax table. Format: string, maxLength=4
TaxableSalesAmount	1	Sum of the taxable value of all sales for this TaxLevelID net of discounts and promotional allowances and after refunds. Must be greater than

		zero. Format: decimal
TaxCollectedAmount	1	The amount of tax collected. Must be greater than zero. Format: decimal
TaxableSalesRefundedAmount	1	Sum of the value of the refunds of the taxable sales for this TaxLevelID. Must be greater than zero. Format: decimal
TaxRefundedAmount	1	The amount of tax refunded. Must be greater than zero. Format: decimal

2.13.17 *TenderInfo*

Data elements that describe a tender.

Tag	Occurs	Description
TenderInfo	1	
TenderCode	1	The PCATS identifier for the tender. This value will be "loyaltyOffer" if the tender represents a transaction-level discount driven by the Loyalty System. Format: as defined by PCATS
TenderSubCode	1	The PCATS subcategory identifier for the tender. Format: as defined by PCATS
ISOPrefix	0..1	The first 6 digits of an account number for card-based methods of payment. The Loyalty System can use this value to identify a card type for the purpose of accruing or redeeming rewards based on specific card types. Format: string, maxLength=6
LoyaltyRewardID	0..1	The identifier of the reward as known to the Loyalty System. Applies only where the tender represents a transaction-level discount driven by the Loyalty System, and where the TenderCode is "loyaltyOffer". See LoyaltyRewardID
TenderAmount	1	The sum of the value of the tender type taken from or given to the Customer. Must be greater than zero. If the ChangeFlag is "yes" then this amount was given to the Customer. Format: decimal
ChangeFlag	1	Indicates if the tender was given as change to the Customer.

	Attribute: value = yes no
--	-----------------------------

2.14 Common Data Element Definitions

Tag	Description
InstantRewardFlag	Attribute: value = yes no If the value attribute is "yes", the reward is applied instantly (forced). If the value is "no", the reward is optional and the POS system must prompt the Customer using the RewardPromptText (Short or Long) to determine if the reward should be redeemed.
LoyaltyID	Attribute: entryMethod = swipe scan RFID manual other A token that represents a customer's unique loyalty identifier. The content of this value is dependent on the entryMethod attribute. An entryMethod of "swipe" implies this value contains the Primary Account Number (PAN). An entryMethod of "scan" implies this value contains a barcode. An entryMethod of "manual" implies this value contains a manually entered account number. Format: string, maxLength=40
LoyaltyRewardID	A unique identifier for a loyalty reward, as known to the Loyalty System. The Loyalty System generates this value for each reward discount provided to the POS System. The POS System must associate this value with the corresponding POS promotion and send this value back to the Loyalty System each time the promotion is referenced in the transaction details. Format: string, maxLength=40
LoyaltySequenceID	A unique identifier for the transaction, as known to the Loyalty System. The Loyalty System generates this value when it sees a new transaction and sends it to the POS System in a response. The POS System must send the same value back to the Loyalty System in all subsequent requests that pertain to the same transaction. Format: string, maxLength=40
POSSequenceID	A unique identifier for the message, as known to the POS System. The POS System can use this value to validate that a response matches up to the original request. The POS System generates this value and sends it to the Loyalty System in a request. The Loyalty System must send the same value back to the POS System in the corresponding response. Format: string, maxLength=40
RewardDiscountMethod	Values: amountOff amountOffPPU

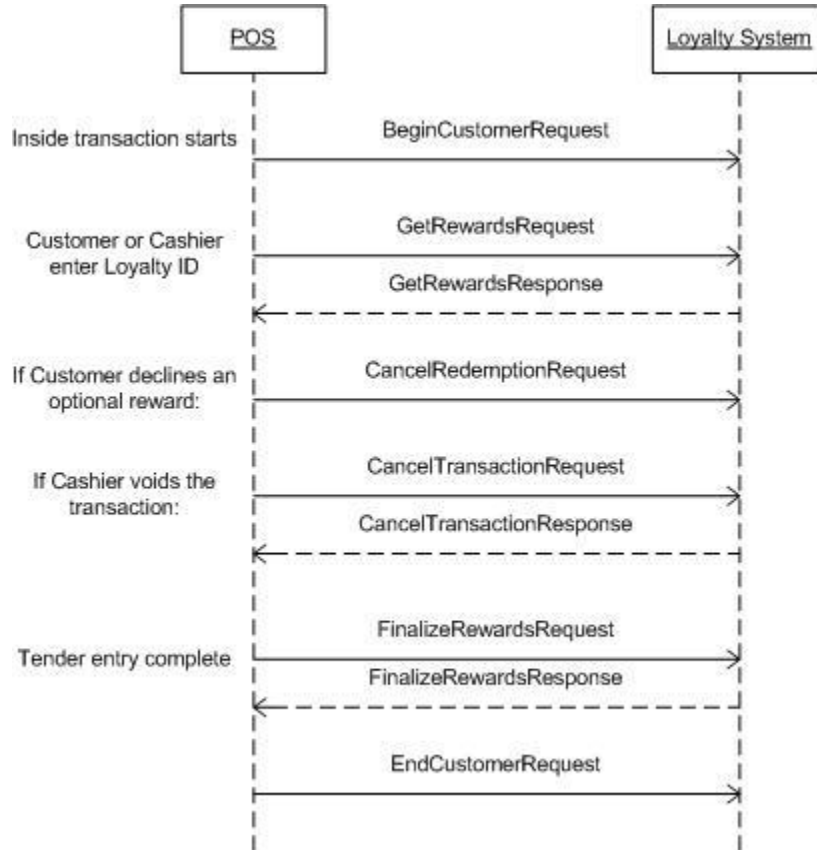
	percentOff newPrice The method used to calculate a discount to be applied when redeeming a reward. For a prepay fuel reward, only amountOffPPU and newPrice apply. For a postpay fuel reward or a non-fuel reward, only amountOff, percentOff and newPrice apply. For a transaction-level reward, only amountOff applies. If an invalid method is specified for the type of reward, the reward is ignored.
RewardLimit	Attribute: type = quantity volume The maximum quantity/volume to which the reward may be applied. Must be greater than zero. If the quantity/volume purchased exceeds this limit, then the reward discount should only be applied to the quantity/volume up to this limit. If this value is not specified then the reward has no limit. This value is ignored for a transaction-level reward. Format: decimal
RewardPromptTextShort	The prompt text to display to the Customer that asks a yes/no question on whether an optional reward should be redeemed. Used for small display devices like pinpads (2x16). If InstantRewardFlag is "no" and this value is empty or not present, the POS should treat the reward as instant. Format: string, maxLength=32
RewardPromptTextLong	The prompt text to display to the Customer or Cashier/Attendant that asks a yes/no question on whether an optional reward should be redeemed. If InstantRewardFlag is "no" and this value is empty or not present, the POS should treat the reward as instant. Format: string, maxLength=80
RewardReceiptDescShort	The text to be used on a 20-column Customer receipt to describe a loyalty reward discount. Format: string, maxLength=8
RewardReceiptDescLong	The text to be used on a 40-column Customer receipt to describe a loyalty reward discount. Format: string, maxLength=24
RewardValue	The value of the reward. Must be greater than zero. For an amountOff, this is the dollar amount to subtract from the unit price. For an amountOffPPU, this is the dollar amount off per gallon. For a percentOff, this is the percent to reduce the unit price, expressed as a decimal value from 0.0 to 1.0. For a newPrice, this is the new unit selling price. The POS System must adjust the reward value if necessary to ensure that no loyalty reward will drive an item's actual sale price below zero.

	Format: decimal
--	-----------------

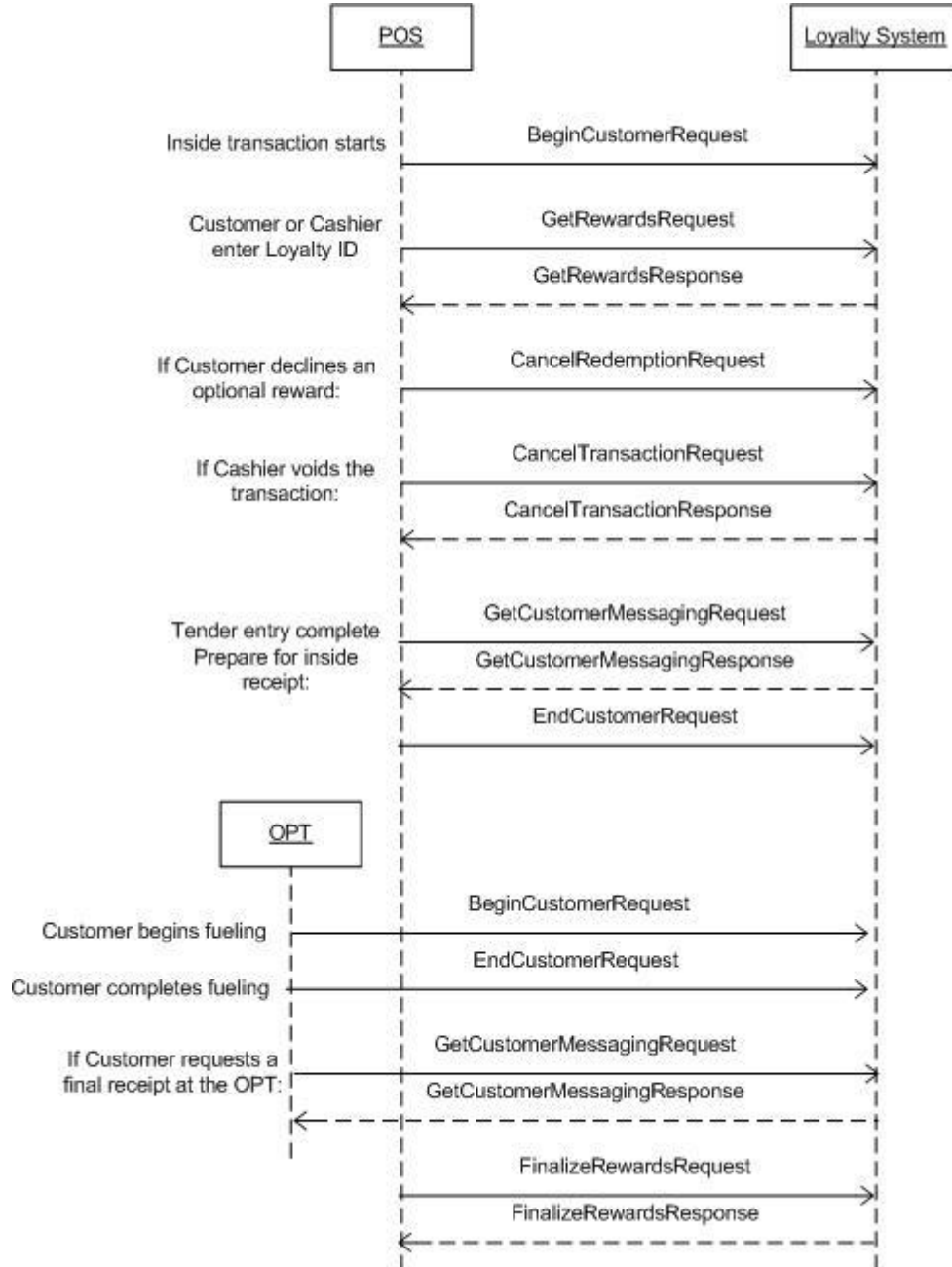
3 Message Flows

The following sections describe the message flows between the POS System and the Loyalty System for each of the major use cases.

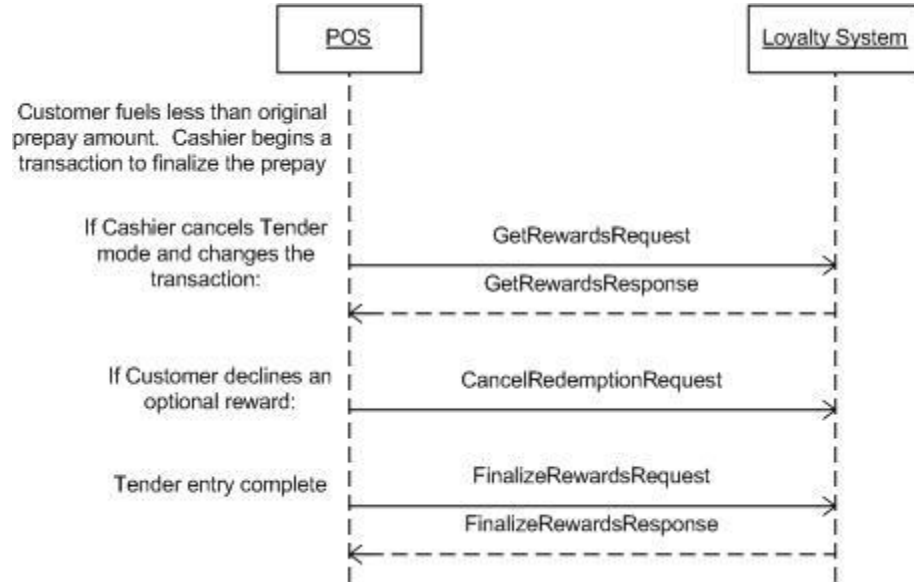
3.1 Inside Sale



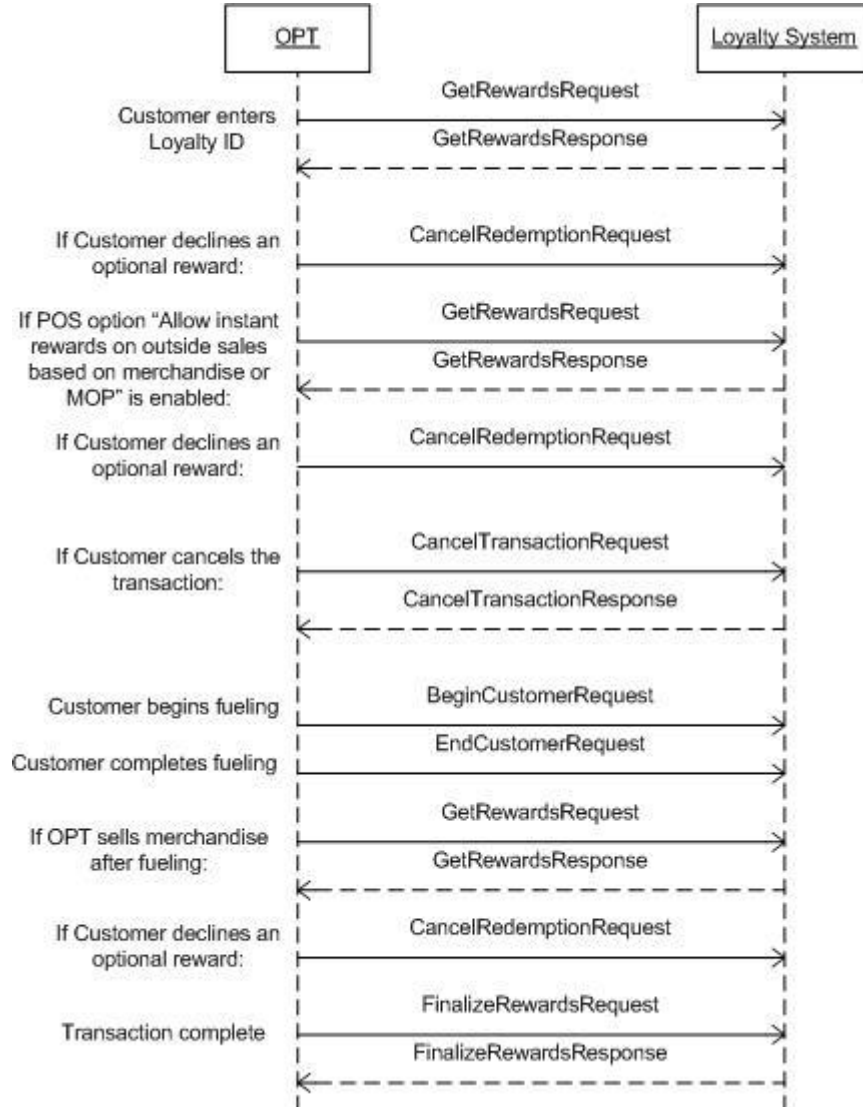
3.2 Prepay Fuel Sale



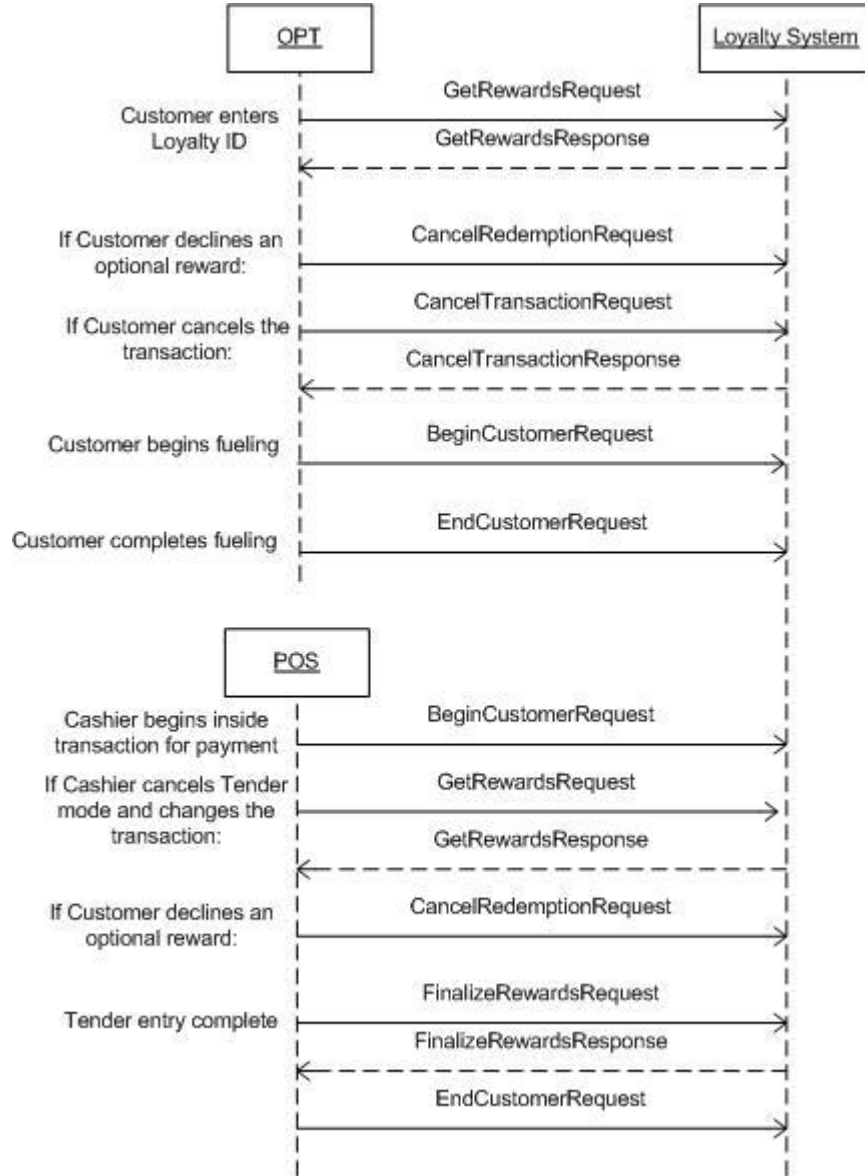
3.3 Prepay Underrun



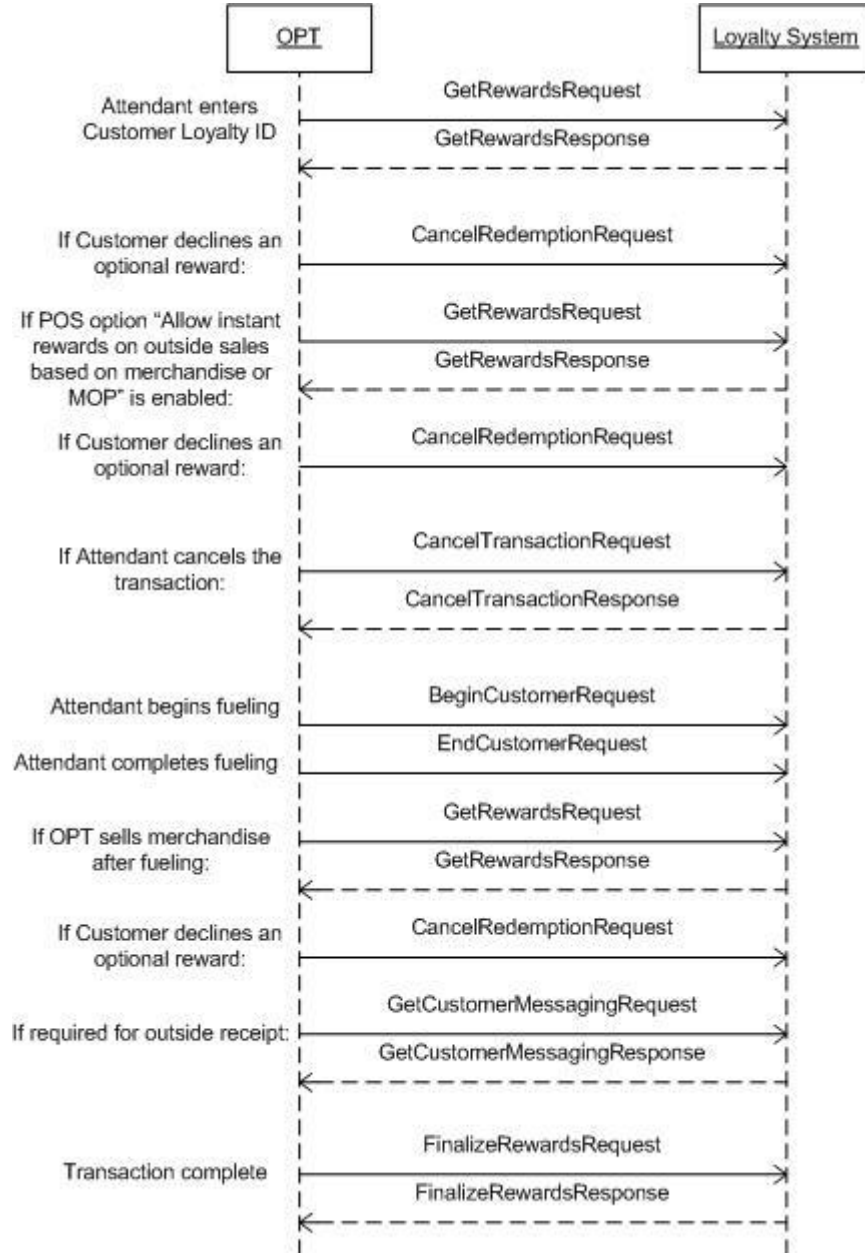
3.4 Outside Sale Paid Outside



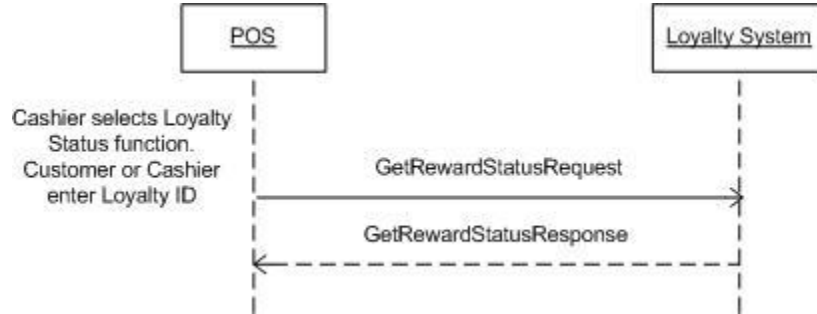
3.5 Outside Sale Paid Inside



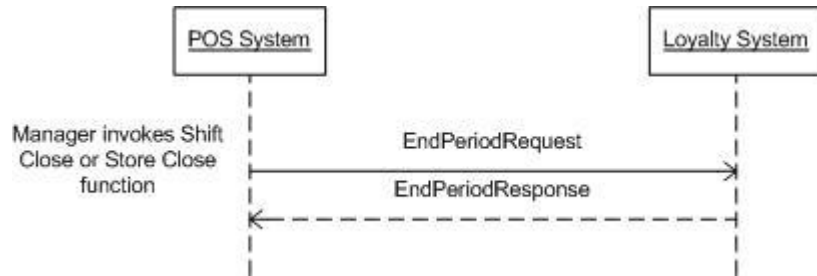
3.6 Outside Attended Sale



3.7 Loyalty Status Request



3.8 End of Period



4 Tag Index

The following index provides a cross reference from each tag name to the page in this document on which the tag is defined.

ActualSalesPrice, 21, 23, 24	GetCustomerMessagingRequest, 13
BeginCustomerRequest, 10	GetCustomerMessagingResponse, 13
BeginDate, 20	GetLoyaltyOnlineStatusRequest, 15
BeginTime, 20	GetLoyaltyOnlineStatusResponse, 16
BusinessDate, 20	GetRewardsRequest, 11
BusinessPeriod, 20	GetRewardsResponse, 11
CancelRedemptionRequest, 13	GetRewardStatusRequest, 14
CancelTransactionRequest, 14	GetRewardStatusResponse, 15
CancelTransactionResponse, 14	InstantRewardFlag, 28
CashierID, 19	ISOPrefix, 27
ChangeFlag, 27	ItemCode, 22
CustomerMessageData, 18	ItemLine, 21
Description, 21, 23, 24	ItemTax, 26
Discount, 26	KioskID, 10
DiscountAmount, 26	LineNumber, 21
DiscountID, 26	LinkedFromLineNumber, 22
DiscountReason, 26	LoyaltyID, 28
DisplayCommand, 18	LoyaltyIDValidFlag, 11
DisplayData, 18	LoyaltyReportLocation, 15
DisplayLine, 18	LoyaltyRewardID, 25, 27, 28
EndCustomerRequest, 10	LoyaltySequenceID, 17, 28
EndDate, 20	MerchandiseCode, 21, 23
EndPeriodRequest, 15	MerchandiseCodeLine, 23
EndPeriodResponse, 15	OutsideSalesFlag, 10, 19
EndTime, 20	POSCode, 22
EntryMethod, 21	POSCodeFormat, 22
ErrorText, 17	POSCodeModifier, 22
EventEndDate, 20	POS LoyaltyInterfaceVersion, 17
EventEndTime, 20	POSSequenceID, 17, 28
EventStartDate, 20	POSTransactionID, 19
EventStartTime, 20	PrepayFuelRewards, 12
FinalizeRewardsRequest, 14	PriceOverride, 23
FinalizeRewardsResponse, 14	PriceOverridePrice, 23
FromPeriod, 15	PriceOverrideReason, 23
FuelGradeID, 11, 24	PriceTierCode, 24
FuelGradeReward, 12	PrimaryReportPeriod, 20
FuelLine, 24	Promotion, 25
FuelPositionID, 10, 24	
FuelPriceData, 11	

PromotionAmount, 25	SellingUnits, 22
PromotionID, 25	ServiceLevelCode, 24
PromotionReason, 25	TaxableSalesAmount, 26
PromptForLoyaltyFlag, 16	TaxableSalesRefundedAmount, 27
ReceiptData, 18	TaxCollectedAmount, 26, 27
ReceiptLine, 18	TaxLevelID, 26
RegisterID, 10	TaxRefundedAmount, 26, 27
RegularSellPrice, 11, 22, 24	TenderAmount, 27
RemoveReward, 13	TenderCode, 27
RequestHeader, 17	TenderInfo, 27
ResponseHeader, 17	TenderSubCode, 27
RewardActions, 11	TillID, 19
RewardDiscountMethod, 28	TimeTierCode, 24
RewardLimit, 29	ToPeriod, 15
RewardPromptTextLong, 29	TrainingModeFlag, 19
RewardPromptTextShort, 29	TransactionData, 19
RewardReceiptDescLong, 29	TransactionDetailGroup, 19
RewardReceiptDescShort, 29	TransactionHeader, 19
RewardTargetLineNumber, 12	TransactionLine, 20
RewardValue, 12, 29	TransactionTax, 26
SalesAmount, 22, 23, 24	VendorModelVersion, 17
SalesQuantity, 22, 23, 24	VendorName, 17
SecondaryReportPeriod, 20	